Improved Reddenings, Distances and Luminosities

for a new sample of Galactic Cepheids

SHUBHAM MAMGAIN

Master Astrophysics

Matr. No. : 794731

SUPERVISOR I - Dr. Jesper Storm

Dwarf Galaxies and the Galactic Halo

Leibniz Institute for Astrophysics Potsdam (AIP)

SUPERVISOR II - Prof. Dr. Maria Rosa Cioni

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Leibniz Institute for Astrophysics Potsdam (AIP)

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Mathematisch-Naturwissenschaftliche Fakultät

Institut für Physik und Astronomie

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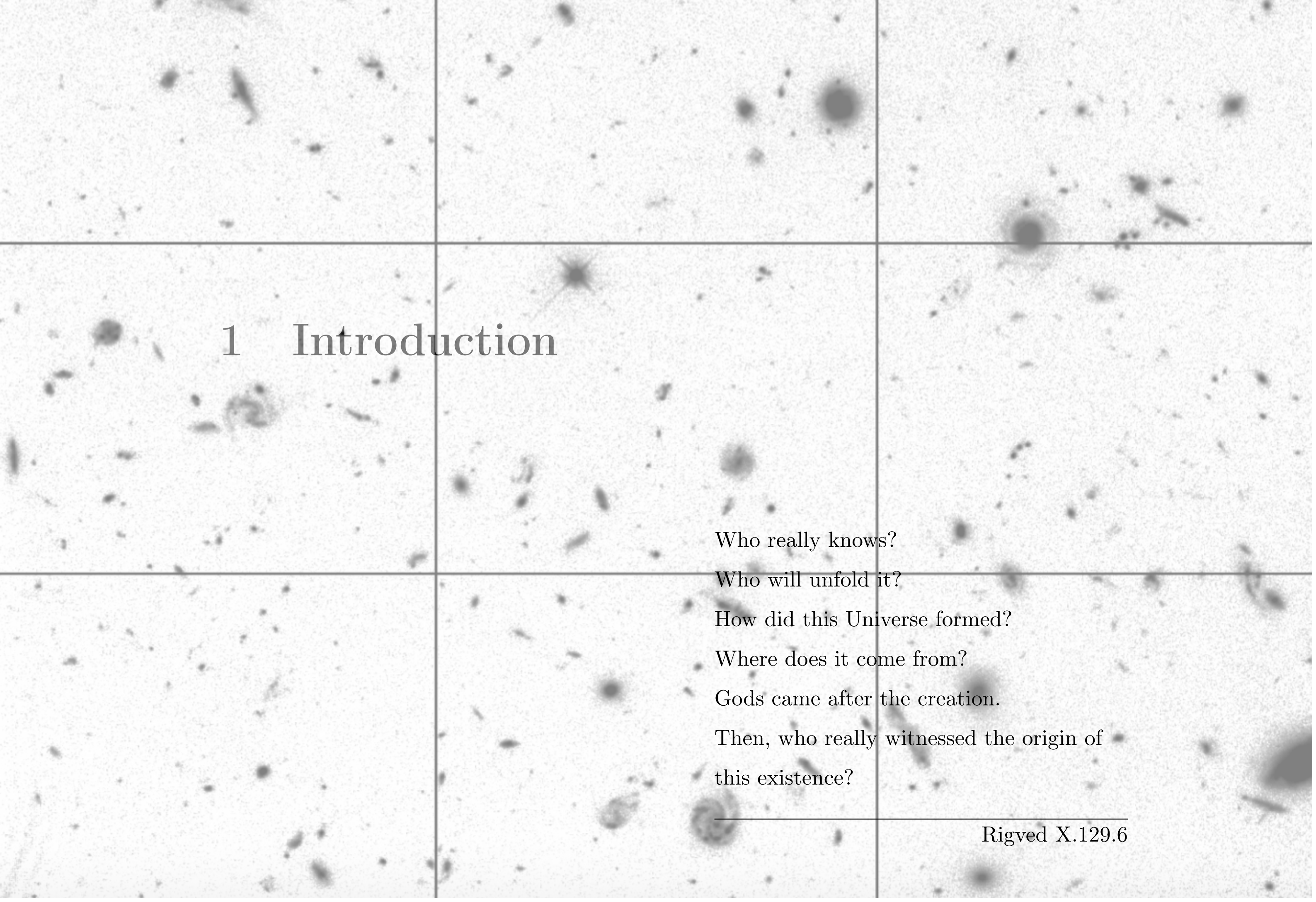

1 Introduction

Who really knows?
Who will unfold it?
How did this Universe formed?
Where does it come from?
Gods came after the creation.
Then, who really witnessed the origin of
this existence?

Rigved X.129.6

1.1 Perspective on the Subject

Have you ever wondered why the Sun shines, why stars exhibit different colors, or what the structure of the Milky Way is? How far is the Andromeda galaxy from Earth? Do structures larger than galaxies exist? How far into the Universe can we observe? And perhaps most intriguingly, are we alone in this vast cosmos?

1.1.1 Astrophysics, Astronomy, and Cosmology

Such fundamental questions have been explored extensively, though many remain only partially answered. By observing the Universe across vast distances and in all directions, and by employing multiple observational messengers—including electromagnetic radiation, neutrinos, and gravitational waves—scientists have begun to unravel several long-standing cosmic mysteries. Astrophysics is inherently interdisciplinary, drawing upon principles from physics, chemistry, geology, computer science, and related fields to construct a coherent understanding of the Universe and humanity's place within it. Astrophysicists investigate the physical processes governing celestial objects, their interactions and evolution, the dynamics of complex cosmic systems, and the potential

[0]Background Image: The Hubble Ultra Deep Field. Credit: Hubble Deep Field Activity (March 2009), NASA.

existence of life beyond Earth.

Astronomy may be regarded as the historical precursor to astrophysics, with a primary emphasis on the observation and systematic description of celestial objects. It focuses on measuring positions, motions, brightness, and classifications, forming the empirical foundation upon which physical interpretations are built. This observational approach is essential for modeling the dynamics of the Universe and for predicting astronomical phenomena such as solar and lunar eclipses, planetary conjunctions, and stellar motions within the Galaxy, as well as for mapping the observable Universe. In this sense, astronomy primarily addresses questions of *what* and *where*, while astrophysics extends this framework to address questions of *how* and *why*.

On the largest scales, cosmology treats the Universe as a single, interconnected physical system. It seeks to understand the origin, geometry, age, evolution, and ultimate fate of the Universe as a whole. Across this hierarchy of spatial scales, galaxies emerge as the fundamental building blocks, which themselves are organized into larger structures such as groups, clusters, and superclusters. These structures are arranged along vast filamentary networks interspersed with low-density regions known as cosmic voids, together forming what is commonly referred to as the cosmic web.

Constrained by observational evidence yet concerned with the most fundamental questions of existence, cosmology occupies a unique position at the intersection of empirical science and philosophical inquiry, and is among the oldest intellectual pursuits of human civilization. Early reflections on the origin and nature of the Universe can be found in ancient texts, including the *Nāsadīya Sūkta* (RV. X.129) of the *Rigveda*, composed in Sanskrit over 2,500 years ago. This hymn offers a speculative meditation on the primordial state of the cosmos, describing an initial condition characterized by darkness, undifferentiated existence, and a generative principle associated with heat or energy preceding the emergence of structure and order.

> There was neither death nor immortality then;
> No distinguishing sign of night nor of day;
> That One breathed, windless, by its own impulse;
> Other than that there was nothing beyond.
>
> Darkness there was at first, by darkness hidden;
> Without distinctive marks, this all was water;
> That which, becoming, by the void was covered;

That One by force of heat came into being.

– *Rigveda* X.129.4–5, translation from Christian (2011)

As an ancient conceptual precursor to modern cosmological thought, such texts reflect a deep philosophical engagement with questions concerning the origin, scale, and structure of the Universe—an inquiry that continues today through increasingly precise observations and theoretical advancements.

1.2 Spectrum of Spatial Scale

In this section, a brief overview of the hierarchical structure of the Universe is presented, as revealed by contemporary astronomical observations.

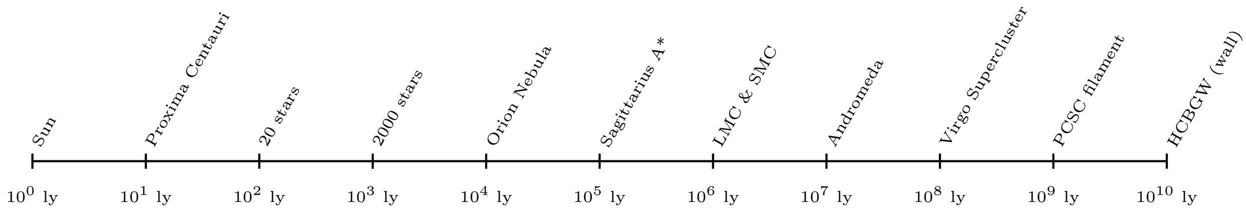

Figure 1.1: Cosmic distances span from nearby stars to the largest known structures in the universe, measured on scales of light-years, typically up to the order of 10. Each label represents either the distance to a notable astronomical object or the size of a vast cosmic region. The diameter of the observable universe is roughly 93 billion light-years, which is nearly the order of 11.

1.2.1 Our Galaxy: Milky Way

To develop an intuition for cosmic scales, it is instructive to compare distances across different astronomical regimes. The nearest star to the Sun, Proxima Centauri, lies at a distance of approximately 4.24 light-years. By definition, one light-year is the distance traveled by light in vacuum over one year, corresponding to about 9.46 trillion kilometers. Within a radius of 10 light-years from the Sun, roughly 20 stars are known, while nearly 2,000 stars reside within 100 light-years. More distant stellar landmarks, such as the stars forming the Orion Belt, are located at distances ranging from approximately 800 to 1,300 light-years.

The Milky Way is a barred spiral galaxy composed of several spiral arms. Its primary spiral arms include the Perseus Arm and the Scutum–Centaurus Arm, while secondary features include the Sagittarius Arm, the Norma Arm, and the Orion Spur. The Solar System is located within the Orion Spur, a minor spiral feature situated between the Perseus Arm and the Scutum–Centaurus Arm, and containing millions of stars.

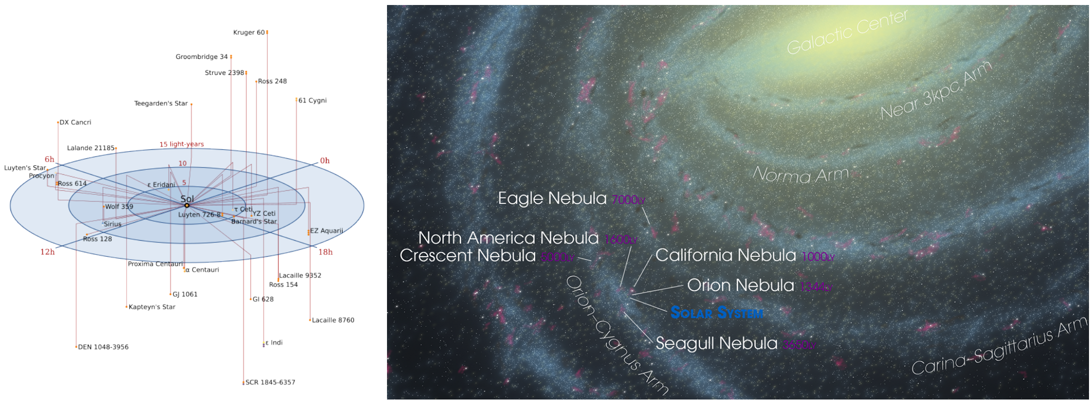

Figure 1.2: *Left:* Stars located within 15ly. *Right:* A segment of the Milky Way depicting the Solar System and its neighboring nebulae located in the Orion–Cygnus Arm. The Galactic Center and other minor arms - Norma and Carina–Sagittarius - are also highlighted. Credit: ESO/R.-D. Scholz et al. (AIP) (left) Andrew Z. Colvin (wikipedia) (Colvin, 2018)(right)

Near the Galactic center, the Perseus and Scutum–Centaurus arms appear to originate close to the ends of the Milky Way's central bar, known as the Long Bar, which extends roughly 15,000–16,000 light-years across the inner Galaxy. At the very center lies Sagittarius A*, a supermassive black hole with a mass of approximately 4 million solar masses, believed to mark the dynamical nucleus of the Milky Way. Sagittarius A* is located at a distance of about 27,000 light-years from Earth. Together, these components define the characteristic structure of the Milky Way: a flattened spiral disk with a central bulge and an extended stellar halo.

1.2.2 Galactic Halo and Surrounding Galaxies

The Milky Way's stellar disk spans approximately 100,000 light-years in diameter and is embedded within an extended, roughly spherical dark matter halo that reaches several times this scale. This halo hosts a population of satellite galaxies, stellar streams, and globular clusters. Among the most prominent satellites is the Large Magellanic Cloud (LMC), the most massive companion of the Milky Way, located at a distance of approximately 160,000 light-years ($\sim$ 50 kpc). The LMC itself possesses a companion, the Small Magellanic Cloud (SMC), situated roughly 200,000 light-years ($\sim$ 60 kpc) from Earth.

To date, more than 50 satellite galaxies of the Milky Way have been identified, ranging from relatively massive systems such as the Magellanic Clouds to ultra-faint dwarf galaxies. The nearest large external galaxy is the Andromeda Galaxy (M31), located at a distance of about 2.54 million light-years ($\sim$ 778 kpc). Andromeda is estimated to be two to three times more massive than the Milky Way and is expected to merge with

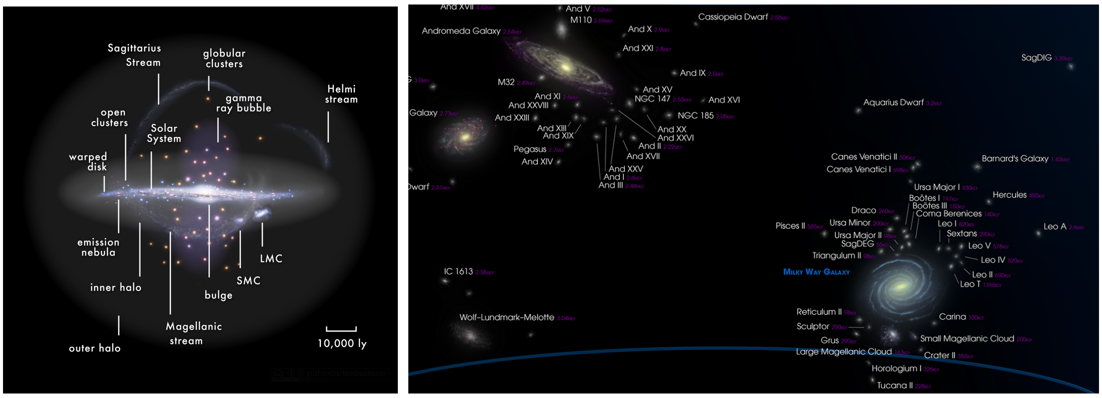

Figure 1.3: *Left:* Side view of the Milky Way, depicting its components and satellite galaxies, notably the Large and Small Magellanic Clouds (LMC and SMC). *Right:* Spatial distribution of galaxies within the Local Group, dominated by the Milky Way and Andromeda galaxies. Source: Wikipedia - Pablo Carlos Budassi (left) and Andrew Z. Colvin (right)

it in several billion years.

Extending the observational horizon to a radius of roughly 10 million light-years reveals a population of several dozen galaxies, dominated by the Milky Way and Andromeda as the most massive members. This ensemble, known as the Local Group, constitutes a gravitationally bound system and occupies a peripheral region of the Virgo Supercluster.

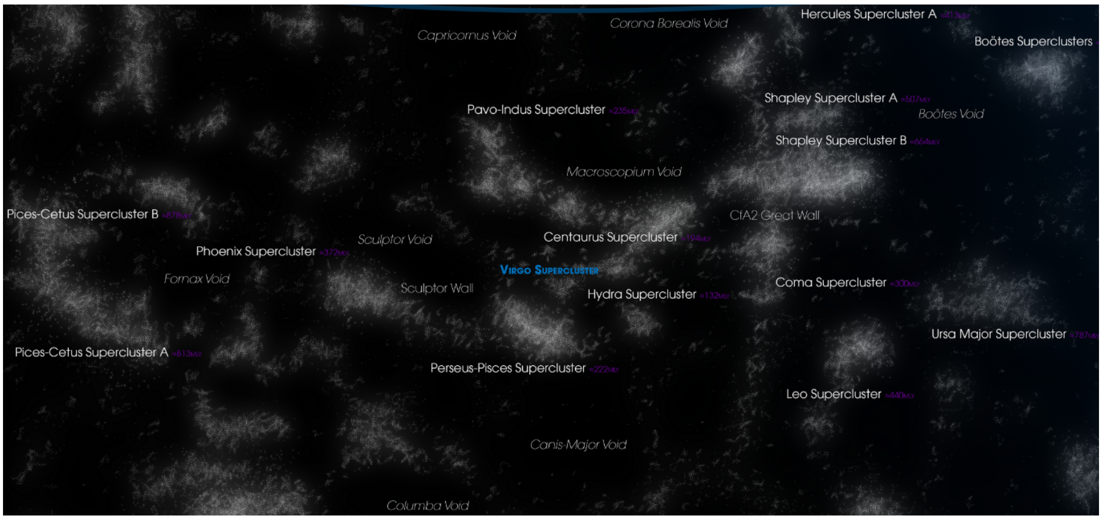

Figure 1.4: Galaxy supercluster and voids within 10^9 ly of Virgo Supercluster. Source: Andrew Z. Colvin (Wikipedia)

1.2.3 Galaxy Clusters and Superclusters

At increasingly larger distances, the megaparsec (Mpc) becomes the standard unit for cosmological measurements, with 1Mpc $\approx$ 3.26 million light-years. The Virgo Supercluster, which includes the Local Group, the Virgo Cluster, and numerous smaller galaxy groups, extends over a scale of approximately 30–35 Mpc (corresponding to $\sim$ 100 million light-years).

The Virgo Supercluster itself is a constituent of the much larger Laniakea Supercluster, a vast gravitational basin encompassing a region several hundred megaparsecs across and containing on the order of 10^5 galaxies. Prominent structures within Laniakea include the Hydra–Centaurus Supercluster, the Pavo–Indus Supercluster, and the Southern Supercluster.

Near the dynamical center of Laniakea, within the Hydra–Centaurus region, lies the so-called Great Attractor—a concentration of mass inferred from galaxy peculiar velocities. Many galaxies within Laniakea, including the Milky Way, exhibit bulk motions directed toward this region under its gravitational influence.

1.2.4 Cosmic Web: Filaments, Walls, and Voids

On the largest observable scales, matter in the Universe is distributed in a complex network known as the cosmic web. Superclusters such as Laniakea, Shapley, Hercules, Coma, and Perseus–Pisces are connected through extended filamentary structures composed of galaxies, gas, and dark matter. One such structure is the Pisces–Cetus Supercluster Complex, a filamentary arrangement estimated to span nearly one billion light-years in length.

Adjacent large-scale features include the Perseus–Pegasus Filament and other interconnected structures of comparable scale. Among the most extensive reported structures is the Hercules–Corona Borealis Great Wall, which may extend over several billion light-years, although its precise size and coherence remain subjects of ongoing investigation.

Together, these filaments, sheets, and vast under-dense regions known as cosmic voids form the cosmic web, which defines the large-scale architecture of the observable Universe. This hierarchical distribution of matter is illustrated schematically in Figure **??**.

1.3 Historical Background and Development

Advances in observational technology have enabled the exploration of the Universe with unprecedented depth and precision. These developments have profoundly influenced modern astronomy and cosmology, providing new insights into the origin, evolution, and large-scale structure of the cosmos. This section presents a concise overview of the historical progression of astronomical research, with particular emphasis on key discoveries and theoretical advances over the past century that have shaped contemporary cosmological thought.

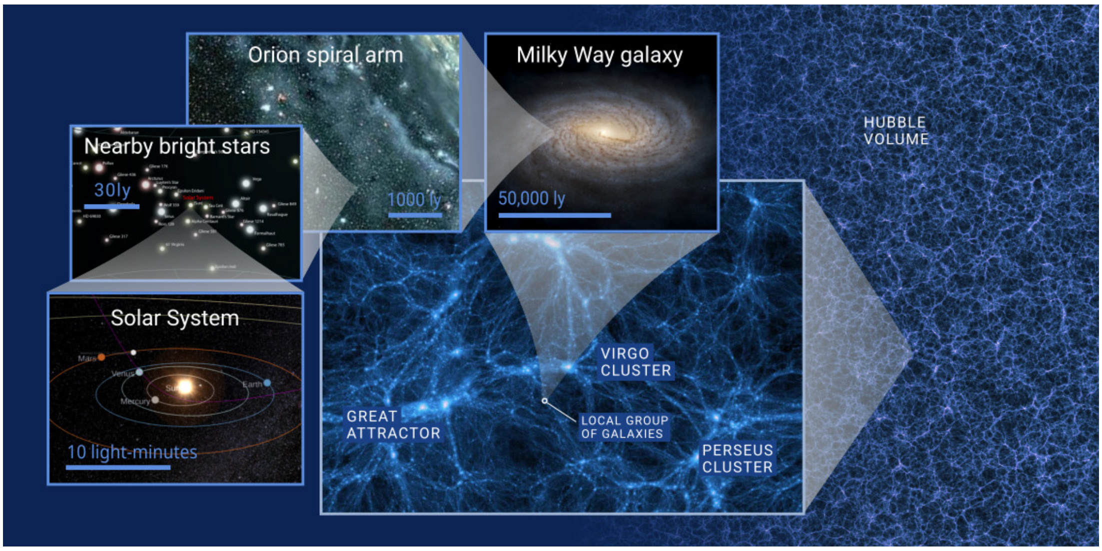

Figure 1.5: Large-scale structure of the Universe illustrating the cosmic web. *Source: S. Stapelberg, Structures blog from the University of Heidelberg*

1.3.1 Distance Determination beyond the Galaxy: Early Twentieth Century

Prior to the twentieth century, the Milky Way was widely believed to constitute the entirety of the Universe. The Sun was assumed to occupy a central position within this system, and the age of the Universe was thought to be far younger than current estimates. Objects now recognized as external galaxies were instead interpreted as nearby gaseous clouds within the Milky Way and were collectively referred to as *spiral nebulae*. This limited understanding of cosmic scale culminated in a prolonged scientific controversy, most notably the Great Debate of 1920 between Harlow Shapley and Heber Curtis.

At the heart of this debate lay the inability to measure reliable distances to spiral nebulae. A decisive breakthrough emerged with the discovery of the period–luminosity relation for pulsating variable stars, specifically Cepheid variables, by Henrietta Swan Leavitt in 1908 and its subsequent refinement in 1912 (Leavitt, 1908; Leavitt and Pickering, 1912). Leavitt demonstrated a tight empirical correlation between the pulsation period of Cepheids and their intrinsic luminosity, establishing that brighter Cepheids exhibit longer periods. In its modern form, this relation may be expressed as

$$M \propto \log P,$$

where M denotes the absolute magnitude and P the pulsation period.

This relation provided, for the first time, a reliable standard candle for estimating distances well beyond the confines of the Milky Way. Its application by Edwin Hubble to Cepheid variables in nearby spiral nebulae led to the unambiguous demonstration that these objects were external galaxies, thereby vastly expanding the known scale of the Universe. The role of the Leavitt Law in establishing the extragalactic distance scale is illustrated in Figure 1.6.

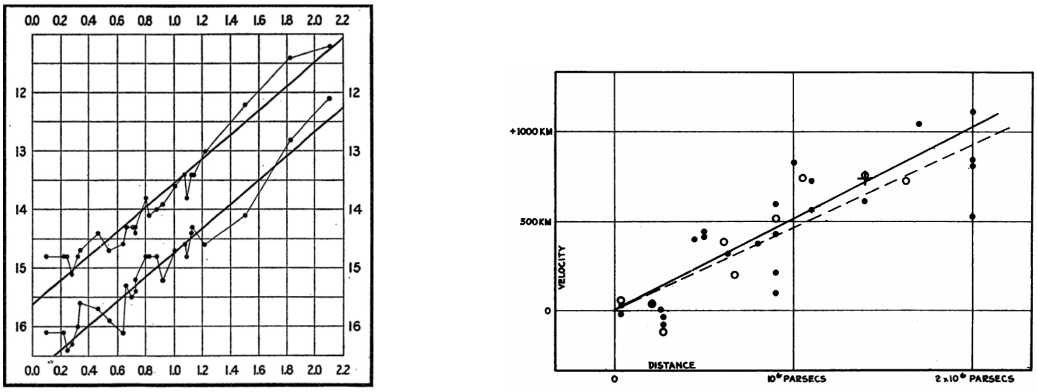

Figure 1.6: Two remarkable discoveries of early twentieth century: a) Leavitt Law: Period (in logarithmic scale) of 25 Cepheids (of Small Magellanic Cloud) correlated with their maximum and minimum brightness. Leavitt and Pickering (1912) b) Hubble Law: Increasing recession velocity of galaxies with respect to distance suggesting an expanding Universe. Hubble (1929)

Note 1.3.1: Knowledge of Distance is fundamental in astronomy.

Distance enables us to convert the sky's seemingly two-dimensional projection into a three-dimensional representation of the Universe. Once the distance to a bright object is known, its true luminosity can be determined, allowing us to infer its size, mass, age and other physical characteristics.

1.3.2 Expansion of the Universe

Einstein's original cosmological model assumed a static and spatially closed Universe. To maintain such a configuration against gravitational collapse, he introduced the cosmological constant, Λ, into the field equations of general relativity (Einstein, 1917). This modification was later rendered unnecessary by observational evidence.

In 1923, using the then state-of-the-art 100-inch (2.5 m) Hooker Telescope at Mount Wilson Observatory, Edwin Hubble observed the Andromeda *spiral nebula* (M31) and resolved individual Cepheid variable stars within it. By applying the period–luminosity relation for Cepheids (Hubble, 1925), Hubble estimated the distance to M31 to be approximately 275 kpc, placing it well beyond the confines of the Milky Way. This result effectively resolved the Shapley–Curtis Great Debate in favor of Heber Curtis, demonstrating that Andromeda is an external galaxy and establishing that the Milky Way is only one among countless galaxies in the Universe.

By extending his observations to additional galaxies, Hubble subsequently uncovered a systematic relationship between distance and radial velocity: galaxies recede from the observer with velocities proportional to their distances, expressed as

$$v = H_0 D,$$

where H_0 is the Hubble constant (Hubble, 1929); see Figure 1.6. This empirical law provided the first direct evidence that the Universe is undergoing a global expansion, fundamentally contradicting Einstein's earlier static model.

The Hubble constant is a cornerstone of modern cosmology, as it sets the present-day expansion rate of the Universe and provides a means to estimate its age and size. Achieving an accurate and precise determination of H_0 remains one of the central objectives of contemporary observational cosmology.

1.3.3 The Big Bang Model: 1930s

Independently of Hubble's observational work, Georges Lemaître applied Einstein's equations of general relativity to cosmology and arrived at solutions describing an expanding Universe. In 1931, he proposed that cosmic expansion originated from an extremely hot and dense initial state, an idea that later evolved into the Big Bang model (Lemaître, 1931).

Further theoretical developments occurred in the late 1940s, when Gamow proposed a hot early Universe model, subsequently refined by Alpher and Herman (Gamow, 1948; Alpher and Herman, 1948). These works predicted the existence of a relic thermal radiation field—the cosmic microwave background (CMB)—as a remnant of the early Universe, with an expected temperature of a few kelvin.

This prediction was observationally confirmed in 1965, when Arno Penzias and Robert Wilson detected an isotropic microwave signal with a temperature of 3.4 ± 1 K using the Holmdel Horn antenna (Penzias and Wilson, 1965). Their discovery provided compelling empirical support for the Big Bang model, which today constitutes the foundation of the standard cosmological paradigm, describing the origin, evolution, and large-scale structure of the Universe.

1.3.4 Composition of Stars

Parallel advances in quantum physics during the early twentieth century profoundly improved the understanding of stellar structure and evolution. In 1917, Arthur Eddington proposed a thermodynamic framework to explain the stability and pulsational behavior of stars, including Cepheid variables, emphasizing the role of internal pressure and energy transport (Eddington, 1917). Building on this work, he famously hypothesized in 1920 that stellar energy is generated through the fusion of hydrogen into helium (Eddington, 1920), a concept that was revolutionary at the time.

Further breakthroughs arose from the analysis of stellar spectra. In her 1925 doctoral dissertation, Cecilia Payne demonstrated that hydrogen and helium are by far the most abundant elements in stellar atmospheres, overturning the prevailing assumption that stellar compositions resembled that of the Earth (Payne-Gaposchkin, 1925). Her work firmly established these elements as the dominant constituents of stars. A comprehensive theoretical description of nuclear energy generation in stars was later provided by Hans Bethe in 1939 (Bethe, 1939), who identified the proton–proton chain and CNO cycle as the primary mechanisms of stellar nuclear fusion.

These developments represented a major turning point in astrophysics, as they established the physical basis of stellar structure and evolution. Prior to this period, the internal mechanisms governing stars—including the origin of the cyclic pulsations observed in Cepheid variables—were largely unknown.

1.3.5 Age of the Universe: 1950s

Through detailed studies of stellar populations, Walter Baade discovered that Cepheid variables comprise two distinct classes: Type I (classical Cepheids), which are young, metal-rich, and intrinsically brighter, and Type II (W Virginis stars), which are older, metal-poor, and less luminous. Baade's recalibration of the Cepheid distance scale led to a doubling of the estimated distance to the Andromeda Galaxy and a corresponding upward revision of the inferred age of the Universe from approximately 1.8 to 3.6 billion years (Baade, 1958).

This revision was further refined by Allan Sandage, who extended the Leavitt relation to incorporate color dependence, yielding a period–luminosity–color (PLC) relation. Sandage also revised the value of the Hubble constant to approximately 75 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (Sandage, 1958), resulting in an estimated cosmic age of order 14 billion years. This value is notably close to modern determinations of the Hubble constant and the age of the Universe.

1.3.6 Technological Development: 1960s

Despite persistent theoretical and observational challenges, technological progress has continually driven advances in astronomy. Beginning in the 1960s, the transition from photographic plates to electronic detectors marked a major transformation in observational capabilities. The introduction of charge-coupled devices (CCDs) in astronomical photometry dramatically improved sensitivity, observation range, and measurement precision Roth (2023). While photographic plates could only detect 1–2 % of incoming photons, CCDs can register up to 90% of incident flux, enabling the observation of much fainter objects. An important advantage of CCD imaging is the digitalization of photometric data. Each CCD chip contains an array of photosensitive pixels, which emits electrons upon absorbing incoming photons via the photoelectric effect Einstein (1905). Higher light intensity leads to higher voltage potential, corresponding to higher pixel values in the resulting area of image. The full grid of pixels digitally maps the segment of the sky exposed to the camera through telescope.

As the light of different frequencies interact with materials differently, CCD imaging technique enable astronomers to observer in wider range of wavelengths. Standard silicon-based CCDs operate efficiently within the optical band. Though, silicon CCDs coated with magnesium fluoride (MgF_2) extend the absorption in UV band Hamden et al. (2011). For Infrared band (700 - 2500 nm), mercury cadmium telluride (HgCdTe) based detectors are used to achieve higher absorptionFox et al. (2009).

On the computational front, sustained advances in high-performance computing have profoundly transformed astronomical research by enabling detailed numerical investigations of complex astrophysical systems. These include simulations of stellar interiors Kippenhahn and Weigert (1990), galaxy formation and evolution Navarro and White (1993), and cosmological structure formation within large-scale simulations Pillepich et al. (2025). Such studies rely on parallel computing and supercomputing architectures to solve coupled gravitational, hydrodynamical, and radiative processes across a wide range of spatial and temporal scales.

The rapid growth of data from ground- and space-based surveys has necessitated advanced infrastructures for data processing, archiving, and dissemination. Initiatives such as the Virtual Observatory (VO) provide standardized, interoperable frameworks for the efficient analysis of distributed astronomical datasets Hatziminaoglou (2009). Complementing these efforts, specialized software tools such as TOPCAT enable efficient visualization, cross-matching, and statistical analysis of large astronomical catalogues while maintaining compatibility with VO standards Taylor (2005). Together, these tools underpin the efficiency, reproducibility, and scientific impact of modern data-driven astronomy.

1.3.7 The Space Age: Gamma, X-Ray, UV, Infrared, Microwave, and Radio Astronomy

The advent of space-based observatories enabled observations across the full electromagnetic spectrum, free from atmospheric absorption, scattering, and turbulence that severely limit ground-based measurements. Early space missions such as the *Orbiting Astronomical Observatory-2* (OAO-2, 1968) and the *International Ultraviolet Explorer* (IUE, 1978) opened access to the ultraviolet (UV) regime, allowing astronomers to probe high-energy stellar and interstellar processes. Observations at even shorter wavelengths became possible with X-ray missions including *Uhuru* (SAS-1, 1970), the *Einstein Observatory* (HEAO-2, 1978), and later the *Chandra X-ray Observatory* (1999). While the first systematic study of high-energy gamma-ray emission began with the *COS-B* mission in 1975.

Among these pioneering space missions, a particularly significant milestone was achieved in astrometry with the launch of the *Hipparcos* (*High Precision Parallax Collecting Satellite*, 1989). Hipparcos measured precise trigonometric parallaxes for approximately 120,000 stars within a distance of about 1,000 light-years (~ 300 pc) Agency (1997). As the first mission dedicated exclusively to high-precision astrometry, Hipparcos enabled the direct calibration of the Leavitt Law using Galactic Cepheid variables, thereby providing a fundamental rung in the cosmic distance ladder Feast and Catchpole (1997).

1.3.8 The Golden Era of Astronomy: Hubble, Planck, Gaia, and JWST

The launch of the *Hubble Space Telescope* (HST) in 1990 marked a pivotal advance in observational astronomy. One of its flagship programs, the *Key Project on the Extragalactic Distance Scale* Freedman et al. (2001), aimed to determine the Hubble constant (H_0) by observing Cepheid variables in galaxies out to ~ 25 Mpc. The decade-long HST Key Project reported

$$H_0 = 72 \pm 8 \text{ km s}^{-1}\,\text{Mpc}^{-1}$$

Leveraging the unprecedented photometric capabilities of the *Hubble Space Telescope*, the *SHOES* (Supernovae, H_0, for the Equation of State of Dark Energy) project calibrated Type Ia supernovae as standard candles Riess and Filippenko (1998). Adam Riess standardized Type Ia supernovae using the Phillips relation Phillips (1993), which correlates the peak luminosity of the light curve with the rate of declining brightness, enabling precise distance measurements to their host galaxies. This methodology extended reliable distance measurements to gigaparsec scales and led to the discovery of the accelerated expansion of the Universe, ushering in the era of precision cosmology Riess and Filippenko (1998).

Complementing local measurements, the *Planck* satellite (2009) conducted an all-sky photometric and polarimetric survey of the cosmic microwave background (CMB), mapping temperature fluctuations and polarization anisotropies that encode information about early-Universe density and velocity fields (Akrami et al., 2020; Planck Collaboration et al., 2020). By fitting these observations to the standard ΛCDM cosmological model, Planck team inferred

$$H_0 = 67.4 \pm 0.5\ \mathrm{km\,s^{-1}\,Mpc^{-1}},$$

The resulting discrepancy between local and early-Universe measurements—termed the *Hubble tension*—currently exceeds 5σ, underscoring the need for improved calibration of the cosmic distance ladder and a deeper understanding of cosmology Riess et al. (2022).

The *Gaia* satellite, launched by ESA in 2013, represents a transformative advance in addressing this tension. Gaia provides microarcsecond-precision astrometry for over 1.8 billion stars upto 10 kiloparsec, yielding accurate parallaxes, proper motions, and a detailed three-dimensional map of the Milky Way Bailer-Jones et al. (2021). These measurements refine the zero-point of the Leavitt Law — a key objective aligned with the focus of this thesis. This ultimately reduces the systematic uncertainties in local distance determinations, directly enabling more precise estimates of H_0 through Cepheid-based calibration

Combined with ongoing HST observations and the deployment of the *James Webb Space Telescope* (JWST, 2021), Gaia firmly establishes the present era as a golden age of astronomy, characterized by multiwavelength coverage, high-precision measurements, and transformative insights into the structure, evolution, and expansion of the Universe.

1.4 Scope of this Thesis

It is evident from the previous sections that accurate distance measurements are essential in astronomy, as they underpin the determination of fundamental physical properties—such as size, mass, and age—across diverse astrophysical and cosmological contexts. Building on this principle, the primary focus of this master's thesis is the calibration of a key distance indicator—the Leavitt Law—by minimizing systematic errors that contribute to scatter in the period–luminosity (PL) relation.

Systematic uncertainties in reddening and distances of Galactic Cepheids directly affect the slope and zero-point of the PL relation. The effect of metallicity is not considered in this thesis. Since the PL relation forms the foundation for the next rung of the cosmic distance ladder, even small errors can propagate and amplify at larger scales. Achieving a precise and well-constrained PL relation is therefore crucial for reliable cosmological measurements. In this thesis, the refinement of the Leavitt Law is implemented through a structured methodology, summarized as follows:

- Identify and remove outliers from the BVIJHK photometric dataset of 150 Galactic Cepheids.
- Refine the Galactic reddening law Fouqué et al. (2007) to minimize deviations between apparent and absolute magnitudes using Wesenheit functions.
- Estimate absolute luminosities of Cepheids using band-wise extinction corrections and distances derived from the Infrared Surface Brightness (IRSB) method Storm et al. (2011) and Gaia parallaxes Clementini et al. (2023).
- Develop and adapt a mathematical framework for PL calibration based on Madore et al. (2017) to improve error corrections.
- Quantify uncertainties in distances and interstellar reddenings for individual Cepheids following the refined methodology.
- Derive the calibrated Leavitt Law using distance- and reddening-corrected absolute magnitudes.
- Cross-validate the calibrated PL relation with Cluster Cepheids **?**.
- Apply the refined Leavitt Law to Cepheids in the Large and Small Magellanic Clouds to determine their distances.

This work is organized into three parts. The first part presents the physics of Cepheid pulsation. The second part develops the calibration methodology for addressing distance and reddening systematics. The final part presents the calibrated Leavitt Law, discusses its implications for the extragalactic distance scale through applications to the Magellanic Clouds, and concludes with prospects for future work.

This research utilizes Virtual Observatory (VO) services for data access, TOPCAT for catalogue analysis, and custom data-processing libraries developed in the Python programming language. The software developed as part of this work is publicly available through a GitHub repository at https://github.com/mshubham00/Leavitt_Law_Calibration.

1.4.1 Part I: Parameters of the Leavitt Law

The period–luminosity (PL) relation depends on two key parameters of Cepheid variables: (a) the pulsation period and (b) the intrinsic (absolute) luminosity. Figure 1.7 illustrates the Leavitt Law in the K band, with distances measured via the Infrared Surface Brightness (IRSB) method (Storm et al., 2011) and interstellar extinction estimated using Fouqué's Galactic extinction law (Fouqué et al., 2007) combined with Sandage estimation of Galactic reddening ratio in visual band (Sandage et al., 2004) and scaled with Fernie's reddening measurements (Fernie, 1994).

Pulsation periods, derived from light curves spanning multiple cycles, are well-determined for all Galactic Cepheids in this study. Stellar light emitted by these stars attenuated by scattering and absorption primarily due to dust and gas along the line of sight. Inaccurate estimation for extinction and distance can introduce significant deviations from the expected linear period–luminosity (PL) relation. Therefore, a central objective of this work is to quantify uncertainties in both distance and reddening for each Cepheid and evaluate their impact on the PL relation.

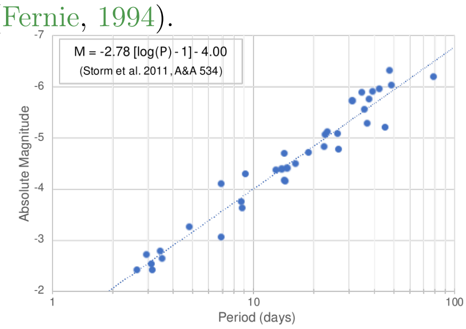

Figure 1.7: *K-band Period-Luminosity relation for Galactic Cepheid using IRSB distance.* Source: Storm J., 2011

This section of the thesis is devoted to examining these parameters, their interdependencies, and the methodologies employed for their determination.

1.4.2 Part II: Calibration Method and Dataset

This part of the thesis describes the Galactic Cepheid dataset, the calibration methodology, discuss underlying physics, and make band-wise analyses of calibrated luminosities. The dataset includes multiband photometry in the BVIJHK bands. Color excesses compiled by Fernie (Fernie, 1994) are converted to band-specific extinction using the Cardelli et al. law (Cardelli et al., 1989) combined with the visual band total-to-selective extinction ratio from Sandage (Sandage et al., 2004). Leavitt Laws are then derived using both Gaia parallaxes and IRSB distances. Prior to calibration, deviations in the slope and intercept of the Leavitt Law for both distance estimates are examined.

The calibration follows Madore's approach (Madore et al., 2017), which employs the reddening-free Wesenheit function (Madore, 1982) to decouple reddening and distance errors. Unlike Madore, who applied a single Wesenheit relation for all residual correlations, this thesis compares residuals from the BVIJHK period–luminosity relations with corresponding composite period–Wesenheit relations. While Madore used only the $V - I$ Wesenheit function, this study extends the method to all fifteen color indices derived from BVIJHK photometry to compute distance–reddening error pairs for each star. Further extending the method to composite Wesenheit functions reveals how period–Wesenheit residuals depend on reddening ratios and color indices.

Finally, the calibrated Leavitt Law is cross-validated using 18 Galactic cluster Cepheids **?**, whose distances are more reliably determined from the averaged parallaxes of their host clusters. This part concludes with an analysis of the calibration results and their implications for refining the PL relation.

1.4.3 Part III: Results and Prospects for Improvement

LMC and SMC Distances

To evaluate the effectiveness of the calibration, distances to the Large and Small Magellanic Clouds (LMC and SMC) are derived using 24 versions of the calibrated Leavitt Law, covering four photometric bands (VIJK) and six color-based Wesenheit functions. This approach allows the construction of a distance matrix that highlights the relative performance of different color indices in producing consistent distance estimates. The results demonstrate the sensitivity of the PL calibration to the choice of photometric band and Wesenheit function, providing guidance for selecting the most robust combination for extragalactic distance determinations.

Effect of Metallicity

While metallicity can affect the zero-point and slope of the period–luminosity relation, it is not explicitly considered in this thesis. However, the framework developed here is readily extendable to incorporate metallicity corrections in future studies, enabling a more comprehensive calibration of the Leavitt Law across different galactic environments.

Target-Based Reddening Law

The refined calibration methodology also provides insights into the reddening law for individual Cepheids. By comparing observed deviations across multiple bands and Wesenheit functions, star-specific reddening ratios can be inferred, allowing a more precise correction of extinction effects. This target-based approach can improve the accuracy of distance estimates, particularly in regions with non-standard or spatially varying extinction, and serves as a foundation for further refinement of both Galactic and extragalactic distance measurements.

Prospects for Further Improvement

The methodology presented here can be extended in several ways:

- Incorporating metallicity corrections to account for chemical composition effects on the PL relation.
- Expanding the dataset to include additional Galactic and extragalactic Cepheids for improved statistical robustness.
- Integrating multi-wavelength data from upcoming surveys (e.g., JWST, LSST) to refine extinction corrections and Wesenheit function formulations.

Together, these improvements will enhance the precision and reliability of the Leavitt Law, thereby reducing systematic uncertainties in the cosmic distance ladder and advancing our understanding of the scale and expansion of the Universe.

Part I

Parameters of Leavitt Law

Part II

Calibration Method and Dataset

Part III

Result Implications

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